

A Generalized Phase Gradient Autofocus Algorithm

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Abstract—The phase gradient autofocus (PGA) algorithm has seen widespread use and success within the synthetic aperture radar (SAR) imaging community. However, its use and success has largely been limited to collection geometries where either the polar format algorithm (PFA) or range migration algorithm is suitable for SAR image formation. In this paper, a generalized phase gradient autofocus (GPGA) algorithm is developed that is applicable with both the PFA and backprojection algorithm (BPA), thereby directly supporting a wide range of collection geometries and SAR imaging modalities. The GPGA algorithm is shown to preserve the four crucial signal processing steps comprising the PGA algorithm while alleviating the constraint of using a single scatterer per range cut for phase error estimation, which exists with the PGA algorithm. Moreover, the GPGA algorithm, whether using the PFA or BPA, is shown to yield an approximate maximum marginal likelihood estimate of phase errors having marginalized over unknown complex-valued reflectivities of selected scatterers. Simulation and experimental results produced by applying the GPGA algorithm to PFA and BPA SAR images are presented.

Index Terms—Phase gradient autofocus (PGA), generalized phase gradient autofocus (GPGA), synthetic aperture radar (SAR) autofocus, backprojection algorithm (BPA), polar format algorithm (PFA), bistatic SAR imaging.

I. INTRODUCTION

A MAJOR challenge faced when performing synthetic aperture radar (SAR) imaging is dealing with phase errors which lead to image defocus. In order to produce high-quality, focused SAR images, accurate models of propagation effects, high-grade transmit and receive hardware, and accurate position and motion measurements of the collection geometry are needed [1]–[3]. The level of accuracy needed is on the order of a fraction of the transmitted signal’s carrier wavelength [1], [2]. Meeting this level of accuracy tends toward costly studies and/or hardware. Typically, a more cost-effective algorithmic approach is considered consisting of a data-driven automatic refocusing technique termed “autofocus” [1]–[3].

Numerous traditional or phase-only autofocus algorithms exist [1], [2], [4]–[50]. Of these many existing autofocus algorithms one is arguably considered the “gold-standard” — the phase gradient autofocus (PGA) algorithm¹ [3], [20]. The PGA algorithm was first introduced in the late 1980s and was formu-

lated for use with the polar format algorithm (PFA) [5]. Since its early development, the PGA algorithm has seen widespread use and success within the SAR imaging community [1]–[3], [5], [8]–[13], [16], [18], [20], [21], [23], [24], [29], [31]–[33], [35], [38], [40], [42], [45], [49]. This success is largely driven by the balance the PGA algorithm provides in terms of being relatively efficient, yet proving to be robust by providing quality autofocus solutions for a number of scene types, scene content, and scenarios. As a result of such widespread use and success, a natural question has been raised in the literature: can the PGA algorithm be applied directly with the backprojection algorithm (BPA)², which directly supports a wide range of collection geometries and SAR imaging modalities [31]? In this work, this question is answered by formulating a generalized phase gradient autofocus (GPGA) algorithm which is applicable with both the PFA and BPA.

In proceeding sections, the GPGA algorithm is formulated beginning with the baseband, received signal model for a bistatic collection geometry (which includes monostatic as a special case). The formulation develops by first providing a side-by-side description of the PGA and GPGA algorithms, making evident the GPGA algorithm includes the PGA algorithm as a special case. Next, the GPGA algorithm is shown to be applicable with the BPA, while preserving the four crucial signal processing steps comprising the PGA algorithm. Lastly, the GPGA algorithm, whether using the PFA or BPA, is shown to provide an approximate maximum marginal likelihood estimate (MMLE) of phase errors having marginalized over unknown complex-valued reflectivities of selected scatterers.

II. NOTATION

The real and complex fields are denoted by $\mathbb{R}$ and $\mathbb{C}$, respectively. The imaginary unit is denoted by $j = \sqrt{-1}$, the complex exponential function by $e : \mathbb{C} \rightarrow \mathbb{C}$, the argument/angle of a complex value by $\text{Arg} : \mathbb{C} \rightarrow \mathbb{R}$, the real operator by $\text{Re} : \mathbb{C} \rightarrow \mathbb{R}$, and the complex conjugate by $(\cdot)^*$. Column vectors are denoted by boldface lowercase letters and matrices by boldface uppercase letters. The n th entry of a vector $\mathbf{x}$ is denoted by $[\mathbf{x}]_n$ using 0-based indexing. The superscripts $(\cdot)^T$ and $(\cdot)^H$ represent the transpose and conjugate transpose, respectively. A square diagonal matrix with main diagonal elements $\mathbf{x}$ is denoted by $\text{Diag}(\mathbf{x})$. The Hadamard or element-wise product is denoted by $\odot$. The vectors $\mathbf{0}$ and $\mathbf{1}$ are vectors of all zeros and ones, respectively, and $\mathbf{I}$ denotes the identity matrix. The Euclidean norm is denoted by $\|\cdot\|$, the ceiling function by $\lceil \cdot \rceil$,

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¹Herein, PGA algorithm refers to the class of PGA algorithms, as variations of PGA exist using different phase estimators [2], [5], [6], [8], [9].

²Herein, BPA refers to the backprojection and filtered/convolution backprojection algorithms.

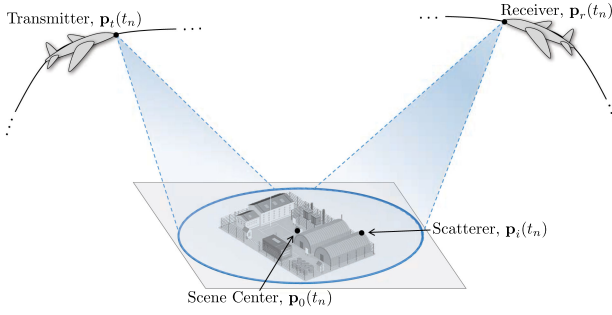


Fig. 1. Spotlight mode bistatic SAR scenario.

and the absolute value of a scalar and determinant of a matrix by $|\cdot|$. Lastly, $\mathcal{U}$ and $\mathcal{CN}$ denote uniform and circularly symmetric complex normal distributions, respectively.

III. PHASE-ONLY AUTOFOCUS SIGNAL MODEL

Consider the spotlight mode bistatic SAR imaging scenario depicted in Fig. 1 consisting of a transmitter-receiver pair and stationary point scatterers. In Fig. 1, $\mathbf{p}_i, \mathbf{p}_t, \mathbf{p}_r : \mathbb{R} \rightarrow \mathbb{R}^3$ are vector-valued functions of time providing the position state of the i th scatterer, transmitter, and receiver, respectively, and $t_n \in \mathbb{R}$ indicates the time of the n th slow-time pulse. Let the true and measured/estimated bistatic delay and Doppler from the transmitter-to- i th scatterer-to-receiver, at slow-time t_n be denoted by $\tau_{i,n}, \tilde{\tau}_{i,n}, \nu_{i,n}, \tilde{\nu}_{i,n} \in \mathbb{R}$, respectively. Moreover, let the n th baseband signal emitted by the emitter be an L -length complex-valued signal denoted by $\mathbf{u}_n \in \mathbb{C}^L$. Then, making stop-and-hop and narrowband signal approximations, the discretized baseband received signal, $\mathbf{s}_n \in \mathbb{C}^L$, for the n th slow-time pulse may be modeled as

$$\mathbf{s}_n = \sum_{m=0}^{M-1} \gamma_{m,n} e^{-j\omega_o \tau_{m,n}} \mathcal{D}(\tau_{m,n}, \nu_{m,n}) \mathbf{u}_n + \mathbf{n}_n, \quad (1)$$

where M is the number of stationary point scatterers, $\gamma_{m,n} \in \mathbb{C}$ is the complex-valued reflectivity for the m th scatterer during the n th slow-time pulse, $\omega_o = 2\pi f_o \in \mathbb{R}$ is the angular carrier frequency of the transmitted signal, $\mathcal{D} : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{C}^{L \times L}$ is the delay-Doppler operator, $\mathbf{n}_n \sim \mathcal{CN}(\mathbf{0}, \sigma_n^2 \mathbf{I})$ is circularly symmetric additive white Gaussian noise (AWGN), and $\sigma_n^2 \in \mathbb{R}$ is the received signal noise power (assumed constant across all slow-time pulses). The unitary delay-Doppler operator for an L -length signal sampled with a sampling rate of f_s is

$$\mathcal{D}(\tau, \nu) = e^{j2\pi\nu\tau} \mathbf{F}^H \text{Diag}(\boldsymbol{\tau}) \mathbf{F} \text{Diag}(\boldsymbol{\nu}), \quad (2)$$

where $\mathbf{F} \in \mathbb{C}^{L \times L}$ is a unitary discrete Fourier transform (DFT) matrix, the ℓ th element of the delay linear phase ramp, $\boldsymbol{\tau} \in \mathbb{C}^L$, is

$$[\boldsymbol{\tau}]_\ell = e^{-j2\pi \frac{\ell}{L} [\ell-L/2]\tau}, \quad (3)$$

and the ℓ th element of the Doppler shift linear phase ramp, $\boldsymbol{\nu} \in \mathbb{C}^L$, is

$$[\boldsymbol{\nu}]_\ell = e^{j2\pi \nu \frac{\ell}{f_s}}. \quad (4)$$

To form a SAR image, the received signal is first range compressed via matched filtering, and phase compensated by the carrier phase term corresponding to the measured bistatic delay to the scene center, $\tilde{\tau}_{0,n}$. Assuming measurement errors are present, the n th range profile or range compressed signal, $\xi_n : \mathbb{R} \rightarrow \mathbb{C}$, at the measured bistatic delay to the i th scatterer, $\tilde{\tau}_{i,n}$, is

$$\begin{aligned} \xi_n(\tilde{\tau}_{i,n}) &= \left(e^{-j\omega_o \tilde{\tau}_{0,n}} \mathcal{D}(\tilde{\tau}_{i,n}, \tilde{\nu}_{0,n}) \mathbf{u}_n \right)^H \mathbf{s}_n \\ &= \gamma_{i,n} e^{-j\omega_o(\tau_{i,n} - \tilde{\tau}_{0,n})} \chi_{i,n}(\tilde{\tau}_{i,n}, \tilde{\nu}_{0,n}) + v_{i,n} + w_{i,n}, \end{aligned} \quad (5)$$

where the matched filtered signal is Doppler compensated to the measured bistatic Doppler to the scene center $\tilde{\nu}_{0,n}$, $\chi_{i,n} : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{C}$ is the n th transmitted signal's ambiguity function whose peak response is located at $(\tau_{i,n}, \nu_{i,n})$, and $v_{i,n}, w_{i,n} \in \mathbb{C}$ are interference and noise terms for the i th scatterer during the n th slow-time pulse, respectively. The interference and noise terms are

$$v_{i,n} = \sum_{\substack{m=0 \\ m \neq i}}^M \gamma_{m,n} e^{-j\omega_o(\tau_{m,n} - \tilde{\tau}_{0,n})} \chi_{m,n}(\tilde{\tau}_{i,n}, \tilde{\nu}_{0,n}), \quad (6)$$

and

$$w_{i,n} = \left(e^{-j\omega_o \tilde{\tau}_{0,n}} \mathcal{D}(\tilde{\tau}_{i,n}, \tilde{\nu}_{0,n}) \mathbf{u}_n \right)^H \mathbf{n}_n, \quad (7)$$

where $w_{i,n} \sim \mathcal{CN}(0, \sigma_w^2)$ and $\sigma_w^2 = \sigma_n^2 \|\mathbf{u}_n\|^2$ is assumed constant across all slow-time pulses (i.e., assume $\|\mathbf{u}_n\|^2 = \|\mathbf{u}_{n'}\|^2$ for $n, n' \in \{0, \dots, N-1\}$).

Upon performing range compression, the next step is to form a SAR image. The SAR image, $\zeta : \mathbb{R}^3 \rightarrow \mathbb{C}$, at location $\mathbf{p}_i$ is

$$\begin{aligned} \zeta(\mathbf{p}_i) &= \tilde{\boldsymbol{\phi}}_i^H \boldsymbol{\xi}_i \\ &= \mathbf{1}^H \tilde{\boldsymbol{\xi}}_i, \end{aligned} \quad (8)$$

where $\tilde{\boldsymbol{\phi}}_i \in \mathbb{C}^N$ is the measured/estimated complex exponential phase vector applied to produce the SAR image at location $\mathbf{p}_i$, $\boldsymbol{\xi}_i \in \mathbb{C}^N$ consists of the N range profile samples contributing to the SAR image at location $\mathbf{p}_i$, and $\tilde{\boldsymbol{\xi}}_i = \boldsymbol{\xi}_i \odot \tilde{\boldsymbol{\phi}}_i^*$ contains the phase compensated range profile samples contributing to the SAR image at location $\mathbf{p}_i$. The expression in Eq. (8) encompasses multiple image formation processes, as each point of the SAR image is expressed as a coherent sum of phase compensated range profile samples. Figure 2 depicts how $\tilde{\boldsymbol{\phi}}_i$, $\boldsymbol{\xi}_i$, and $\tilde{\boldsymbol{\xi}}_i$ are defined for two of the most popular image formation processes: 1) PFA and 2) BPA. For the PFA, the phase vector $\tilde{\boldsymbol{\phi}}_i$ is a Fourier basis vector, and $\boldsymbol{\xi}_i$ are the range profile samples obtained after resampling the phase history data from a polar to a uniformly sampled Cartesian grid. For the BPA, the n th element of the phase vector is $[\tilde{\boldsymbol{\phi}}_i]_n = e^{-j\omega_o(\tilde{\tau}_{i,n} - \tilde{\tau}_{0,n})}$, and the n th element of the range profile samples is $[\boldsymbol{\xi}_i]_n = \xi_n(\tilde{\tau}_{i,n})$.

In the proceeding paragraphs, the expressions introduced for the BPA are used to build the phase-only autofocus signal model. It is important to note that using the BPA for analysis does not preclude the PFA, since loosely speaking, the PFA is a special

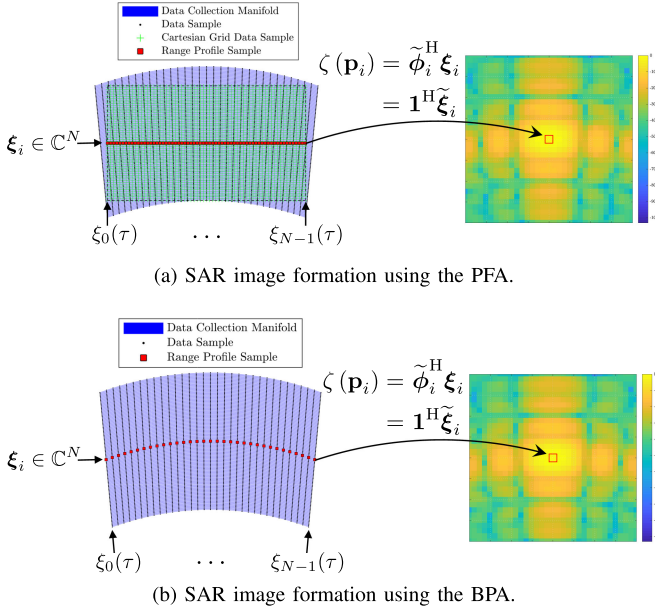


Fig. 2. SAR image formation processes.

case of the BPA. In this special case, the stand-off ranges of the transmitter and receiver to the scene center approaches infinity such that the data collection manifold maps out a uniformly sampled Cartesian grid. Hence, the need for performing polar-to-Cartesian resampling of the phase history data is alleviated (i.e., there is zero wavefront curvature such that the received data already lies on a uniformly sampled Cartesian grid).

Using the expressions from Eqs. (5) to (8) for the BPA, the n th phase compensated range profile sample contributing to the SAR image at location $\mathbf{p}_i$ is

$$[\tilde{\xi}_i]_n = e^{j\omega_o \delta\tau_{i,n}} (\gamma_{i,n} \chi_{i,n}(\tilde{\tau}_{i,n}, \tilde{\nu}_{0,n}) + v_{i,n}) + w_{i,n}, \quad (9)$$

where $\delta\tau_{i,n} = \tilde{\tau}_{i,n} - \tau_{i,n}$ is the measurement error for the bistatic delay to the i th scatterer during the n th slow-time pulse, and the interference and noise terms after phase compensation (i.e., $\tilde{\xi}_i = \xi_i \odot \tilde{\phi}_i^*$) are redefined as

$$\begin{aligned} v_{i,n} &:= e^{-j\omega_o \delta\tau_{i,n}} [\tilde{\phi}_i]_n^* v_{i,n} \\ &= \sum_{\substack{m=0 \\ m \neq i}}^M \gamma_{m,n} e^{j\omega_o (\tau_{i,n} - \tau_{m,n})} \chi_{m,n}(\tilde{\tau}_{i,n}, \tilde{\nu}_{0,n}), \end{aligned} \quad (10)$$

and

$$\begin{aligned} w_{i,n} &:= [\tilde{\phi}_i]_n^* w_{i,n} \\ &= \left(e^{-j\omega_o \tilde{\tau}_{i,n}} \mathcal{D}(\tilde{\tau}_{i,n}, \tilde{\nu}_{0,n}) \mathbf{u}_n \right)^H \mathbf{n}_n. \end{aligned} \quad (11)$$

Given the expression in Eq. (9), the vector of phase compensated range profile samples contributing to the SAR image at location $\mathbf{p}_i$ is

$$\tilde{\xi}_i = \text{Diag}(\delta\phi) \gamma_i + \mathbf{w}_i, \quad (12)$$

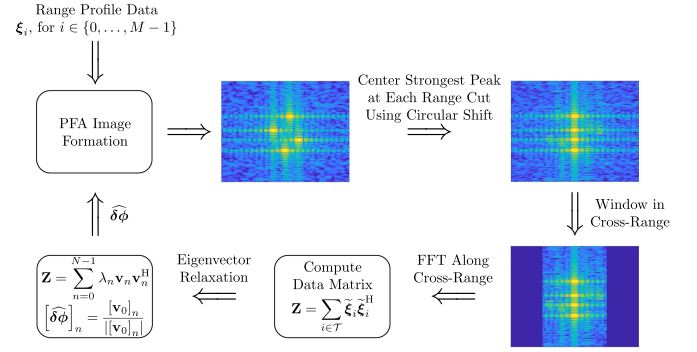


Fig. 3. PGA algorithm from [8].

where each of the terms is defined as follows:

$$[\delta\phi]_n = e^{j\omega_o \delta\tau_{i,n}}, \quad (13)$$

$$[\gamma_i]_n = \gamma_{i,n} \chi_{i,n}(\tilde{\tau}_{i,n}, \tilde{\nu}_{0,n}) + v_{i,n}, \quad (14)$$

$$[\mathbf{w}_i]_n = w_{i,n}. \quad (15)$$

In Eq. (13), the phase error vector is spatially variant as the quantity depends upon i (i.e., the phase error vector changes for each location $\mathbf{p}_i$ in the SAR image). However, in [51] it is shown that for appropriate SAR scene sizes the spatially variant phase error vector is approximately spatially invariant (i.e., the phase error vector is approximately independent of the location $\mathbf{p}_i$ in the SAR image for reasonable SAR scene sizes). Herein, it is assumed an appropriate SAR scene size is chosen so the spatially invariant phase error model is applicable. Hence, a subscript i is omitted from the phase error vector $\delta\phi$ in Eq. (13).

The model in Eq. (12) is the phase-only autofocus signal model assumed for traditional or phase-only autofocus algorithms. In the next section, the PGA algorithm as described in [8], is discussed and a generalization of the algorithm is presented which is applicable with both the PFA and BPA.

IV. GENERALIZED PHASE GRADIENT AUTOFOCUS

In the proceeding discussion, the GPGA algorithm is developed and discussed. The development proceeds by first providing a side-by-side description of the PGA and GPGA algorithms, making evident the GPGA algorithm includes the PGA algorithm as a special case. Next, discussion follows on how the GPGA algorithm applies with the BPA. Then, common strategies for applying the GPGA algorithm are discussed. The section concludes with a discussion on the impact and usefulness of the GPGA algorithm.

A. GPGA Using the PFA

For all remaining discussion, unless explicitly stated, the PGA algorithm refers to the algorithm depicted in Fig. 3, which consists of processing steps described in [8]. Before discussing each of these processing steps, it is first important to understand the assumed relationship between the range profile or range compressed data and the SAR image.

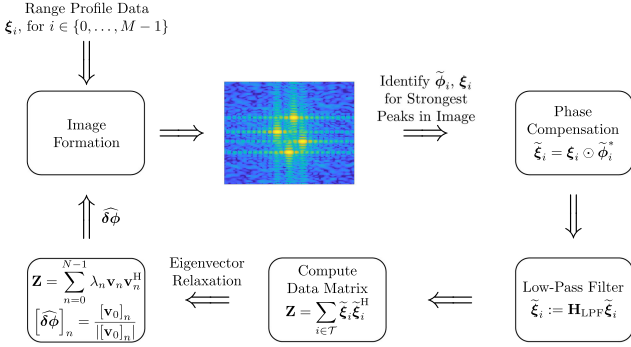


Fig. 4. GPGA algorithm.

In [8], the PGA algorithm is formulated using the PFA, meaning the range profile data is related to the SAR image by a fast Fourier transform (FFT) across slow-time. As alluded to in Section III with Eq. (12), this implies $\tilde{\phi}_i$ is a Fourier basis vector, and ξ_i is a single row of range profile samples from the polar-to-Cartesian resampled data. As will be seen, understanding this relationship is important as well-known Fourier identities are used to develop the GPGA algorithm.

The PGA algorithm depicted in Fig. 3 begins by first forming a PFA image. Next, the strongest peaks from a subset of range cuts from the formed PFA image are centered in cross-range using a circular shift. The set of indices corresponding to the identified strongest peaks (i.e., selected scatterers) is denoted by $\mathcal{T}$. Upon selecting and centering scatterers, a window is applied along the cross-range dimension for each selected range cut. Then, an FFT is computed along cross-range to move from the image domain to the range profile domain. After moving to the range profile domain, what is referred to as the data matrix is computed as a sum of outer products of the circularly shifted and windowed range cuts. Lastly, phase errors are estimated using eigenvector relaxation (EVR), which selects and normalizes the eigenvector corresponding to the dominant eigenvalue of the data matrix³ (i.e., $\mathbf{v}_0$ in Fig. 3).

Figure 4 depicts what is termed the GPGA algorithm. Using Fourier identities, the GPGA algorithm performs each of the signal processing steps of the PGA algorithm directly in the range profile domain. For instance, centering the strongest peak in a single range cut using a circular shift in the image domain is equivalent to element-wise multiplying the range profile samples by the phase vector applied to produce the strongest peak.

³By the relationship between the eigen and singular value decomposition (SVD), the leading eigenvector of the data matrix is equivalent to the leading singular vector of the collection of vectors $\tilde{\xi} = \{\tilde{\xi}_i \mid i \in \mathcal{T}\}$. The complexity for computing the leading singular vector of $\tilde{\xi} \in \mathbb{C}^{N \times P}$ is $\mathcal{O}(NP)$, whereas the complexity for computing the leading eigenvector of $\mathbf{Z} \in \mathbb{C}^{N \times N}$ is $\mathcal{O}(N^2)$ [52]. Thus, when $P < N$, a condition which may be enforced by the system designer, EVR may be performed more efficiently by calculating the leading singular vector of $\tilde{\xi}$ instead of the leading eigenvector of $\mathbf{Z}$.

Mathematically, this equivalence is

$$\begin{array}{c} \text{Center } \zeta(\mathbf{p}_i) \\ \text{Using Circular Shift} \end{array} \iff \tilde{\xi}_i = \xi_i \odot \tilde{\phi}_i^* \quad (16)$$

Applying a window along the cross-range dimension in the image domain is equivalent to applying a low-pass filter (LPF) to phase compensated range profile samples. Mathematically, this equivalence is

$$\begin{array}{c} \text{Window in} \\ \text{Cross-Range} \end{array} \iff \tilde{\xi}_i := \mathbf{H}_{\text{LPF}} \tilde{\xi}_i, \quad (17)$$

where $\mathbf{H}_{\text{LPF}} \in \mathbb{C}^{N \times N}$ is an LPF matrix. The remaining steps of the PGA algorithm, which include computing the data matrix and EVR³, are performed in the range profile domain and are identical for the GPGA algorithm. Thus, it is seen that each of the signal processing steps comprising the PGA algorithm may be performed directly in the range profile domain by way of the GPGA algorithm depicted in Fig. 4.

At this point in the discussion, the GPGA algorithm has been described as being applied with the PFA to enable a side-by-side comparison to be made with the PGA algorithm. In the next section, the GPGA algorithm is described as being applied with the BPA.

B. GPGA Using the BPA

As depicted in Fig. 4, the GPGA algorithm using the BPA begins by first forming a BPA image. Next, the range profile samples and phase vector corresponding to the strongest peaks in the formed BPA image are identified. These quantities are ξ_i and $\tilde{\phi}_i$ for $i \in \mathcal{T}$, where again, $\mathcal{T}$ is the set of indices corresponding to the strongest peaks (i.e., selected scatterers) in the formed BPA image. For the BPA, the range profile samples ξ_i need not correspond to a single range cut (see Fig. 2b), and the phase vector need not correspond to a Fourier basis vector (recall, the BPA phase vector is $[\tilde{\phi}_i]_n = e^{-j\omega_o(\tilde{r}_{i,n} - \tilde{r}_{0,n})}$). However, the centering step is still completed through phase compensation, where an element-wise multiply $\tilde{\xi}_i = \xi_i \odot \tilde{\phi}_i^*$ is computed. Upon performing phase compensation, the windowing step is performed by applying a LPF to the phase compensated range profile samples, as shown in the right hand side of Eq. (17). Next, what is referred to as the data matrix is computed as a sum of outer products of the phase compensated and low-pass filtered range profile samples $\tilde{\xi}_i$. Lastly, phase errors are estimated using EVR, which selects and normalizes the eigenvector corresponding to the dominant eigenvalue of the data matrix³ (i.e., $\mathbf{v}_0$ in Fig. 4).

Before discussing strategies for applying the GPGA algorithm, it is first important to point out some of the intuition behind the centering and windowing steps of the GPGA algorithm when used with the BPA (this same intuition for the centering and windowing steps of the PGA algorithm when used with the PFA is discussed in [2], [9].) First, phase compensation is performed to effectively center the point spread function of each selected scatterer at the scene center of the SAR image (i.e., at the spatial location $\mathbf{p}_0$ where the differential bistatic delay is zero). This centering of the point spread function is better

evidenced by expanding Eq. (8)

$$\begin{aligned}\zeta(\mathbf{p}_i) &= \mathbf{1}^H \tilde{\boldsymbol{\xi}}_i \\ &= \sum_{n=0}^{N-1} \left[\tilde{\boldsymbol{\xi}}_i \right]_n e^{j\omega_o(\tilde{\tau}_{0,n} - \tilde{\tau}_{0,n})} \\ &= \tilde{\phi}_0^H \tilde{\boldsymbol{\xi}}_i,\end{aligned}\quad (18)$$

which closely resembles the expression for the SAR image at the scene center: $\zeta(\mathbf{p}_0) = \tilde{\phi}_0^H \tilde{\boldsymbol{\xi}}_0 = \mathbf{1}^H \tilde{\boldsymbol{\xi}}_0$. Upon centering/aligning the point spread function of each selected scatterer, windowing is performed. Windowing is performed to try to capture the majority of the blur or defocus of each point spread function, while filtering energy outside of the main defocus region to try to improve the effective signal-to-interference-plus-noise ratio (SINR) of each selected scatterer [2], [9].

From [2], [9] and the discussion points above, one may see that the GPGA algorithm when used with the BPA shares the same intuitions as the PGA and GPGA algorithms when used with the PFA. Albeit, the intuition for the steps performed with the BPA is not as readily seen as when the PFA is used, where well-known Fourier identities may be leveraged. In proceeding sections, strategies for performing the four crucial signal processing steps of the GPGA algorithm are discussed.

C. Scatterer Selection

As will be discussed and shown in Section V, with the GPGA algorithm there is statistical motivation for selecting scatterers with either moderate or nearly constant SINR. Since these scattering locations are not generally known *a priori*, a strategy must be devised for identifying these scattering locations from formed (and potentially defocused) SAR images.

Arguably, the most straightforward approach for scatterer selection is to apply a relative threshold to the formed SAR image. That is, indices of image points with an image intensity within a specified threshold of the maximum image intensity are included in $\mathcal{T}$. The intuition for this approach is that image points with higher intensity potentially have higher SINR.

A potentially more robust and complicated scheme may consist of using strategies from constant false alarm rate (CFAR) detection. For instance, the 2D CFAR windowing/filtering process could be applied to the formed SAR image to estimate the SINR at each point in the image. Then, image points with an estimated SINR exceeding a specified SINR threshold could be included in $\mathcal{T}$. For more details regarding the many 2D CFAR windowing/filtering strategies available, the reader is referred to [53]–[56].

D. Low-Pass Filter Design

For GPGA LPF design two strategies commonly employed with the PGA algorithm are proposed [2]. These strategies include: 1) setting the low-pass width to be the estimated blur width of the phase errors and 2) employing an LPF with a constant shrinking low-pass width over GPGA iterations.

For estimating the blur width of the phase errors, a non-coherent sum of the spectrum of each set of phase compensated range profile samples is computed as

$$\mathbf{x} = \sum_{i \in \mathcal{T}} \left(\mathbf{F} \tilde{\boldsymbol{\xi}}_i \right) \odot \left(\mathbf{F} \tilde{\boldsymbol{\xi}}_i \right)^* . \quad (19)$$

Substituting in Eq. (12) for $\tilde{\boldsymbol{\xi}}_i$, and assuming nearly constant reflectivity, high SINR scatterers are selected during the scatterer selection process (i.e., $\gamma_i \approx \gamma_i \mathbf{1}$ and $\mathbf{w}_i$ is negligible), $\mathbf{x}$ from Eq. (19) is seen to be an estimate of the magnitude of the spectrum of the phase errors

$$\begin{aligned}\mathbf{x} &= \sum_{i \in \mathcal{T}} \left(\mathbf{F} (\text{Diag}(\boldsymbol{\delta}\phi) \gamma_i + \mathbf{w}_i) \right) \\ &\quad \odot \left(\mathbf{F} (\text{Diag}(\boldsymbol{\delta}\phi) \gamma_i + \mathbf{w}_i) \right)^* \\ &\approx \sum_{i \in \mathcal{T}} |\gamma_i|^2 \left(\mathbf{F} \boldsymbol{\delta}\phi \right) \odot \left(\mathbf{F} \boldsymbol{\delta}\phi \right)^* .\end{aligned}\quad (20)$$

Thus, from $\mathbf{x}$ the blur or low-pass width of the window in cross-range, let it be denoted by W in terms of the number of samples, may be estimated by thresholding $\mathbf{x}$ at a chosen level below the maximum value of $\mathbf{x}$ (similar to the approach suggested in [2]). Once the width has been estimated, the LPF matrix used in Eq. (17) may be computed by

$$\mathbf{H}_{\text{LPF}} = \mathbf{F}^H \text{Diag}(\mathbf{h}) \mathbf{F}, \quad (21)$$

where

$$[\mathbf{h}]_n = \begin{cases} 1, & |n - N/2| \leq \frac{W}{2}, \\ 0, & \text{else,} \end{cases} \quad (22)$$

for $n \in \{0, \dots, N-1\}$.

The second and simpler GPGA LPF design strategy is to employ an LPF with a constant shrinking low-pass width over GPGA iterations. The intuition here is that after each iteration the image becomes better focused, and the residual phase errors approach a constant. Hence, the spectrum of the residual phase errors has less blur width (i.e., as the residual phase errors approach a constant, the spectrum approaches a delta function with no blur width). For the constant shrinking low-pass width approach, the LPF matrix may be computed using Eq. (21) with $\mathbf{h}$ defined as

$$[\mathbf{h}]_n = \begin{cases} 1, & |n - N/2| \leq \alpha^k \frac{N}{2}, \\ 0, & \text{else,} \end{cases} \quad (23)$$

where $\alpha \in (0, 1]$ is the constant shrink factor and $k \in \{0, 1, \dots\}$ is the GPGA iteration number.

E. Phase Estimation

Similar to the PGA algorithm in [2], [5], [6], [8], [9], the GPGA algorithm supports a class of different phase estimators (although the discussion herein follows similarly to that in [8] where the EVR phase estimator is the focus of the discussion). In other words, the GPGA algorithm is not restricted to performing EVR phase estimation. Rather, the algorithm may be applied with other popular phase estimators such as the phase difference estimator discussed in [2], [9]. For example, using the notation

presented herein, the phase difference estimator from [2], [9] for $n \in \{0, \dots, N-1\}$ is

$$\text{Arg} \left(\left[\widehat{\delta\phi} \right]_n \right) = \begin{cases} 0, & n = 0, \\ \text{Arg} \left(\sum_{n'=0}^{n-1} \sum_{i \in \mathcal{T}} \left[\tilde{\xi}_i \right]_{n'}^* \left[\tilde{\xi}_i \right]_{n'+1} \right), & \text{else.} \end{cases} \quad (24)$$

For more details regarding the relationship between many of the different phase estimators available, the reader is referred to [5], [6], [8], [9].

F. Iterations

As discussed in [8], [9], a crucial step of the PGA algorithm is iterating the algorithm. The intuition for this step is that with each iteration the SAR image becomes better focused enabling the strongest scattering locations to more easily be identified, and thereby leads to improved phase estimation. As depicted in Fig. 4, the GPGA algorithm is designed to preserve the crucial step of iterating.

In [2], [9] strategies or stopping criteria are discussed for determining an appropriate number of PGA algorithm iterations. These strategies apply equally well with the GPGA algorithm. The simplest of these strategies, and the strategy used for results presented in Section VI, is running the GPGA algorithm for a fixed number of iterations. Similar to findings in [8] with the PGA algorithm, the authors have found the GPGA algorithm using EVR phase estimation provides adequate performance with a number of data sets using no more than three GPGA iterations. However, the system designer is free to change the iteration strategy or stopping criteria as seen fit, so both a robust and computationally efficient implementation of the GPGA algorithm may be realized.

G. Summary

At first glance, the GPGA algorithm may only seem useful for applying the PGA algorithm with the BPA. However, the generalization also has an implication which improves upon the PGA algorithm from [2], [5], [6], [8], [9] using the PFA. For instance, with the PGA algorithm there is a known limitation in which only the strongest peak per range cut may be centered and used for phase error estimation [3], [32], [42]. This limitation is alleviated using the generalized implementation shown in Fig. 4, as the phase compensation and low-pass filtering steps may be performed on any arbitrary set of image points $\mathcal{T}$.

In the next section, the GPGA algorithm, whether using the PFA or BPA, is shown to provide an approximate maximum marginal likelihood estimate (MMLE) of the phase errors having marginalized over unknown complex-valued reflectivities of selected scatterers.

V. MAXIMUM MARGINAL LIKELIHOOD ESTIMATOR

Let the variables in Eqs. (13) to (15) be distributed as

$$\omega_o \delta \tau_{i,n} \sim \mathcal{U}[-\pi, \pi), \quad (25)$$

$$\gamma_i \sim \mathcal{CN}(\mathbf{0}, \sigma_{\gamma,i}^2 \mathbf{1}\mathbf{1}^H + \sigma_{v,i}^2 \mathbf{I}), \quad (26)$$

$$\mathbf{w}_i \sim \mathcal{CN}(\mathbf{0}, \sigma_w^2 \mathbf{I}), \quad (27)$$

where $\sigma_{\gamma,i}^2 \in \mathbb{R}$ is the expected matched filtered power per pulse for the i th scatterer and $\sigma_{v,i}^2, \sigma_w^2 \in \mathbb{R}$ are the expected matched filtered interference and noise powers per pulse at the i th scattering location, respectively. In Eqs. (25) to (27) each of the variables are assumed to be independent with respect to one another. For the statistical model described by Eqs. (25) to (27), the following is observed:

- 1) The complex-valued reflectivities for the i th scatterer, $\gamma_{i,n}$, are expected to be perfectly correlated from pulse-to-pulse (i.e., perfectly correlated across n).
- 2) Both the matched filtered interference and noise powers, $\sigma_{v,i}^2$ and σ_w^2 , are constant from pulse-to-pulse (i.e., constant across n).

From the observations stated above a physical implication follows. In particular, for an anisotropic scattering model a small enough receive aperture, in terms of angular extent, must be considered for performing autofocus such that the scatterer's complex-valued reflectivities and the interference power are expected to be constant from pulse-to-pulse. This physical implication does not represent a limitation or shortcoming of an autofocus algorithm leveraging the above statistical model. Rather, this physical implication dictates how an autofocus algorithm leveraging the above statistical model must be applied. For instance, if a wide aperture collection geometry is considered, then the wide aperture data should be broken up into smaller sub-apertures (a step commonly done in practice) for performing autofocus so that the statistical model from Eqs. (25) to (27) applies.

The SINR, given the above statistical model, for the i th scattering location is

$$\text{SINR}_i = \frac{\sigma_{\gamma,i}^2}{\sigma_{v,i}^2 + \sigma_w^2}. \quad (28)$$

Without loss of generality, let the interference-plus-noise term be one (i.e., $\sigma_{v,i}^2 + \sigma_w^2 = 1$), so SINR is parameterized solely by the expected scatterer's matched filtered power $\sigma_{\gamma,i}^2$. Using this assumption and the derivation from Appendix A, the marginal likelihood for the phase compensated range profile samples from Eq. (12), having marginalized over the unknown complex-valued reflectivities γ_i , is

$$\tilde{\xi}_i | \delta\phi, \boldsymbol{\eta} \sim \mathcal{CN}(\mathbf{0}, \sigma_{\gamma,i}^2 \delta\phi \delta\phi^H + \mathbf{I}), \quad (29)$$

where $\boldsymbol{\eta} \in \mathbb{R}^P$ is a vector containing all $\sigma_{\gamma,i}^2$ for $i \in \mathcal{T}$ (i.e., vector containing all hyperparameters) and P is the number of scatterers selected during the scatterer selection process (i.e., the cardinality of $\mathcal{T}$).

Using the matrix determinant lemma and Woodbury matrix identity, the determinant and inverse of the covariance matrix from Eq. (29) are

$$|\sigma_{\gamma,i}^2 \delta\phi \delta\phi^H + \mathbf{I}| = 1 + \sigma_{\gamma,i}^2 N, \quad (30)$$

and

$$(\sigma_{\gamma,i}^2 \delta\phi \delta\phi^H + \mathbf{I})^{-1} = \mathbf{I} - \frac{\delta\phi \delta\phi^H}{\frac{1}{\sigma_{\gamma,i}^2} + N}. \quad (31)$$

Substituting Eqs. (30) and (31) in the log marginal likelihood function for Eq. (29) produces

$$\begin{aligned} \mathcal{L}(\delta\phi, \eta) &\propto - \sum_{i \in \mathcal{T}} \ln(1 + \sigma_{\gamma,i}^2 N) + \sum_{i \in \mathcal{T}} \frac{|\tilde{\xi}_i^H \delta\phi|^2}{\frac{1}{\sigma_{\gamma,i}^2} + N} \\ &\approx - \sum_{i \in \mathcal{T}} \ln(1 + \sigma_{\gamma,i}^2 N) + \alpha(\eta) \sum_{i \in \mathcal{T}} |\tilde{\xi}_i^H \delta\phi|^2, \end{aligned} \quad (32)$$

where constants with respect to $\delta\phi$ and η have been omitted and $\alpha : \mathbb{R}^P \rightarrow \mathbb{R}$, $\alpha(\eta) > 0$ is an approximation constant. For the approximation in Eq. (32) to hold, it is assumed selected scatterers have moderate or nearly constant SINR (i.e., $N\sigma_{\gamma,i}^2 \gg 1$ or $\sigma_{\gamma,i}^2$ is nearly constant for all $i \in \mathcal{T}$). One of these two conditions will typically be satisfied using the GPGA algorithm. In fact, in many cases both conditions will be satisfied. For instance, the first step of the GPGA algorithm is the selection of strongest peaks from the formed SAR image (i.e., image points with potentially high SINR). Moreover, the system designer may enforce that the image intensities of selected scatterers lie within a given interval such that the image intensities are approximately constant (i.e., image points with potentially constant SINR).

From Eq. (32) the approximate log marginal likelihood is seen to be separable in terms of the phase errors and the hyperparameters

$$\mathcal{L}(\delta\phi, \eta) \approx \mathcal{L}_1(\eta) + \alpha(\eta) \mathcal{L}_2(\delta\phi), \quad (33)$$

where $\mathcal{L}_1 : \mathbb{R}^P \rightarrow \mathbb{R}$ and $\mathcal{L}_2 : \mathbb{C}^N \rightarrow \mathbb{R}$ are defined as

$$\mathcal{L}_1(\eta) = - \sum_{i \in \mathcal{T}} \ln(1 + \sigma_{\gamma,i}^2 N), \quad (34)$$

$$\mathcal{L}_2(\delta\phi) = \sum_{i \in \mathcal{T}} |\tilde{\xi}_i^H \delta\phi|^2. \quad (35)$$

Thus, assuming moderate or nearly constant SINR scatterers are selected, an MLE for the phase errors is obtained by

$$\begin{aligned} \hat{\delta\phi} &= \arg \max_{\delta\phi \in \mathcal{P}} \mathcal{L}_2(\delta\phi) \\ &= \arg \max_{\delta\phi \in \mathcal{P}} \sum_{i \in \mathcal{T}} |\tilde{\xi}_i^H \delta\phi|^2 \\ &= \arg \max_{\delta\phi \in \mathcal{P}} \delta\phi^H \mathbf{Z} \delta\phi, \end{aligned} \quad (36)$$

where $\mathcal{P}$ is the set of N -length complex-valued vectors with unit-modulus elements (i.e., set of N -length phase vectors). The GPGA phase estimation problem in Eq. (36) is non-deterministic polynomial-time hard (NP-hard) [38], [42], [57]. For SAR imaging applications, where the number of slow-time pulses is typically large, it is not reasonable to expect to always

solve Eq. (36) (i.e., find a global maximum point). Thus, heuristics are needed to obtain an approximate solution to Eq. (36) in a reasonably efficient manner. The GPGA algorithm, as described in Section IV, obtains an approximate solution to Eq. (36) using EVR (although, as discussed in Section IV other phase estimators may be employed with the GPGA algorithm). The EVR phase estimator selects and normalizes the eigenvector corresponding to the dominant eigenvalue of the data matrix³ $\mathbf{Z}$. For additional details regarding the EVR phase estimator, the reader is referred to [8], [38], [42].

In light of the many comments made in the literature regarding assumptions and shortcomings of the PGA algorithm, the authors feel it is necessary to highlight assumptions made and address potential shortcomings with the GPGA algorithm:

- 1) In Eq. (1), the SAR scene is assumed to consist of a collection of stationary point scatterers — a common modeling assumption made with many SAR imaging applications. Moreover, Eq. (1) relies on stop-and-hop and narrowband signal approximations, and best models scenarios where scattering bodies are large compared to the carrier wavelength [58], [59].
- 2) Timing/position errors, which may result from propagation effects, hardware limitations, and/or motion measurement errors, are assumed to produce spatially invariant phase errors. For this assumption to hold, the SAR scene size is assumed to be chosen according to the scene size limits derived in [51].
- 3) For the statistical model described by Eqs. (25) to (27), the GPGA algorithm provides an approximate MMLE of phase errors having marginalized over unknown complex-valued reflectivities of selected scatterers. For the presented formulation of the GPGA algorithm, no assumptions regarding specific scene types or scene content are made (e.g., scene containing strong isolated point scatterers and/or low/high contrast regions). This statement is not to be misinterpreted as reading, “the GPGA algorithm performs well with all different scene types,” just that the formulation does not rely upon any scene type or scene content assumptions.
- 4) Naturally, the performance of the GPGA algorithm (as well as the PGA algorithm) depends upon the SINR of selected scatterers. Thus, for SAR scenes containing moderate to high SINR scatterers and/or regions, it is reasonable to assume the GPGA algorithm will provide an adequate autofocus solution (although, extreme edge cases may exist where this is not true). For scenes devoid of these convenient moderate or high SINR features, the GPGA algorithm phase error estimation accuracy is generally expected to degrade. Depending upon a number of factors, most notably SINR, this reduction in phase error estimation accuracy may or may not be suitable. In other words, while the estimation accuracy of the GPGA algorithm is generally expected to degrade with SAR scenes containing lower SINR scatterers and/or regions, it is difficult to say precisely for which scene types and scene content the

GPGA algorithm phase error estimation accuracy will be inadequate.

In the next section, example results are presented demonstrating the effectiveness of the GPGA algorithm using both the PFA and BPA for SAR image formation.

VI. RESULTS

In this section, four SAR autofocus results produced using the GPGA algorithm are presented. First, Monte Carlo simulation results are presented where the mean squared error (MSE) of the argument of GPGA phase error estimates are evaluated versus SINR, and compared to the Cramér-Rao lower bound (CRLB). For the second result, the GPGA algorithm is used to autofocus a measured, near-field bistatic SAR data set collected using a small-scale experimental setup in the Radar Instrumentation Laboratory (RAIL) at the Air Force Institute of Technology (AFIT). For the third and fourth results, the GPGA algorithm is applied to measured SAR data from the Gotcha Public Release Dataset [60], where synthetic phase errors are applied to the data to produce image defocus. In the third result, the GPGA algorithm is applied with the PFA. With the fourth result, the GPGA algorithm is applied with the BPA.

For each of the presented autofocus results, the GPGA algorithm (as depicted in Fig. 4) is applied and configured as follows: 1) to attempt to select either moderate or constant SINR scatterers, only image points with an image intensity within 10 dB of the maximum image intensity may be selected (i.e., a relative threshold of 10 dB is applied as described in Section IV-C); 2) a constant low-pass width shrinking factor of $\alpha = 0.8$ is used for LPF design with Eqs. (21) and (23); 3) the GPGA algorithm is run for three iterations. Each of these settings are free to be changed by the system designer as seen fit. However, the authors have found the listed configuration works well on a number of data sets.

A. Monte Carlo Simulation Result

Using the derivation in [8], the inverse of the $(N - 1)$ -by- $(N - 1)$ Fisher information matrix⁴ for the argument of the phase errors, $[\theta]_n = \text{Arg}([\delta\phi]_n)$ for $n \in \{0, \dots, N - 2\}$, is

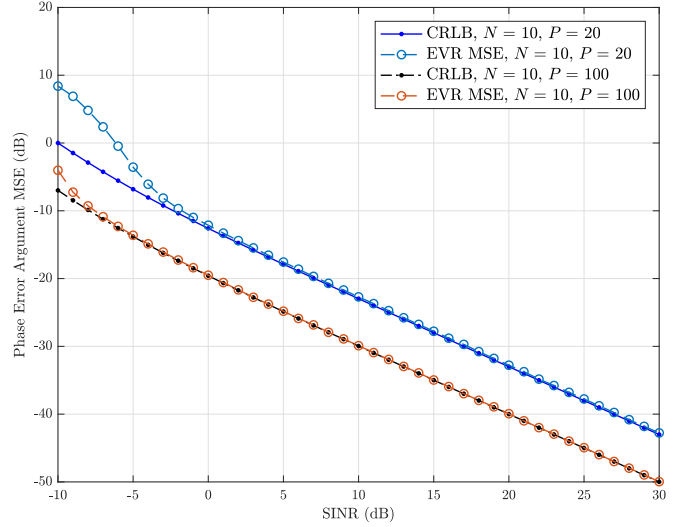
$$\mathbf{J}^{-1} = \frac{1}{2N \sum_{i \in \mathcal{T}} \frac{(\sigma_{\gamma,i}^2)^2}{1 + N\sigma_{\gamma,i}^2}} (\mathbf{I} + \mathbf{1}\mathbf{1}^H). \quad (37)$$

Assuming the expected matched filtered power for each selected scatterer is the same (i.e., $\sigma_{\gamma,i}^2$ is constant for $i \in \mathcal{T}$), the CRLB for the argument of the phase errors is

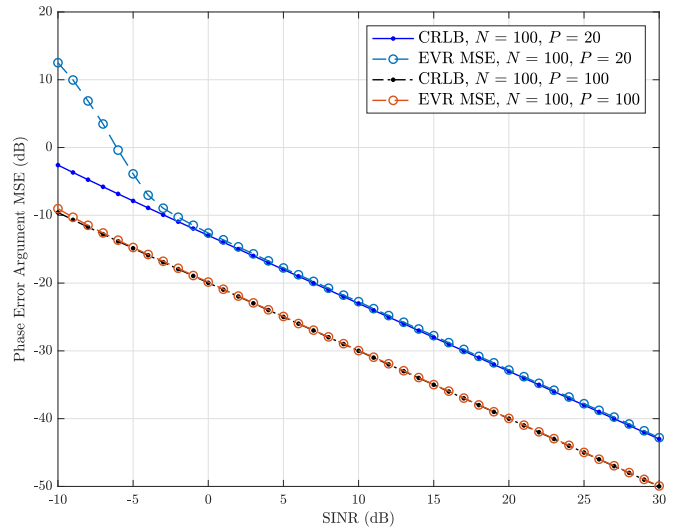
$$[\mathbf{J}^{-1}]_{n,n} = \frac{1 + N\sigma_{\gamma,i}^2}{NP(\sigma_{\gamma,i}^2)^2}, \quad (38)$$

where again, P is the cardinality of $\mathcal{T}$. From Eq. (38) it is seen that for large pulse-SINR products (i.e., $N\sigma_{\gamma,i}^2 \gg 1$), the CRLB

⁴The Fisher information matrix is only $(N - 1)$ -by- $(N - 1)$ as one element of the phase error estimate may be arbitrarily set and all other elements expressed relative to that element (i.e., only $(N - 1)$ values need to be estimated since the autofocus solution is invariant to a constant phase offset).



(a) $N = 10, P \in \{20, 100\}$.



(b) $N = 100, P \in \{20, 100\}$.

Fig. 5. MSE of the argument of EVR phase error estimates versus SINR, for different number of pulses, N , and selected scatterers, P .

is inversely related to only the SINR and number of selected scatterers. In other words, for large pulse-SINR products the CRLB is approximately

$$[\mathbf{J}^{-1}]_{n,n} \approx \frac{1}{P\sigma_{\gamma,i}^2}. \quad (39)$$

Thus, for an efficient estimator with a large pulse-SINR product, the variance on phase error estimates is driven primarily by the SINR and number of selected scatterers. This relationship provides statistical motivation for a key processing step of the GPGA algorithm—the selection of strongest peaks from formed SAR images.

Figure 5 depicts Monte Carlo simulation results showing the MSE of the argument of EVR phase error estimates versus SINR for different number of pulses and selected scatterers (i.e., N and P). In addition, Fig. 5 shows the CRLB for each SINR

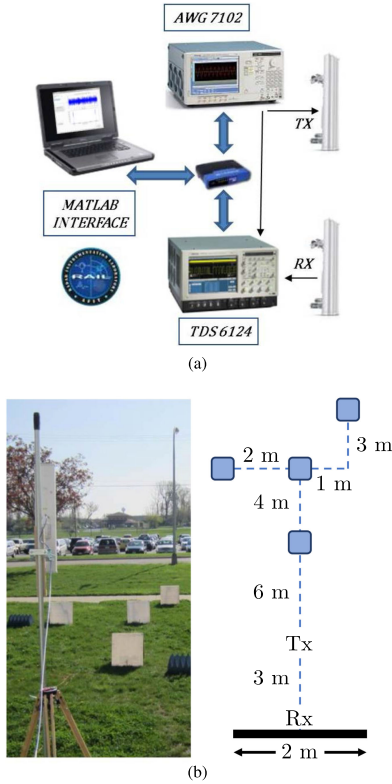


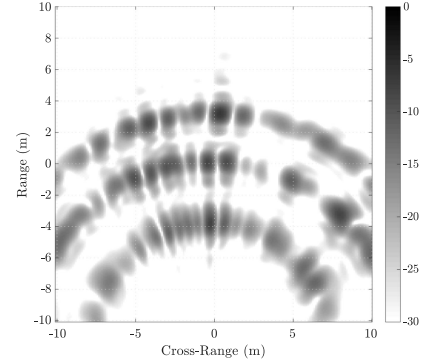
Fig. 6. Experimental system and setup from [47], [61]–[64].

calculated from Eq. (38). For each trial of the Monte Carlo simulation, P vectors containing constant SINR range profile samples are drawn from the marginal distribution in Eq. (29), and used with the EVR phase estimation procedure to produce a phase error estimate. Then, the MSE of the argument of the EVR phase error estimates, compared to the argument of the true phase errors, is computed. For each SINR shown in Fig. 5, 30×10^3 Monte Carlo trials are run.

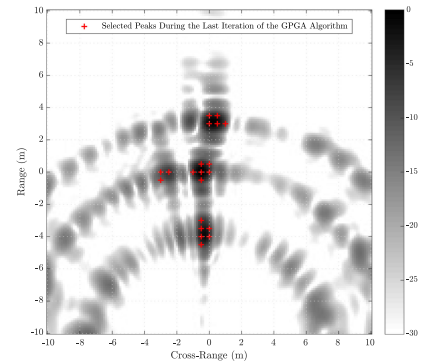
From Figs. 5a and 5b it is evident that for large pulse-SINR products the MSE of the argument of phase error estimates is approximately independent of the number of pulses N . In particular, in Figs. 5a and 5b the MSE and CRLB approach nearly the same value as SINR increases, whether using $N = 10$ or $N = 100$ (e.g., as SINR increases, each of the results in Fig. 5a approach nearly the same value as the results in Fig. 5b).

B. RAIL SAR Data Using the BPA

Figure 6 depicts the small-scale experimental setup used for collecting bistatic SAR data in the RAIL at AFIT. The system and setup shown in Fig. 6 has been used previously to generate test data for passive and bistatic SAR algorithm validation [47], [61]–[64]. On the transmit side, the RAIL system consists of a Tektronix arbitrary waveform generator (AWG) connected to an AirMax 2G-16-90 sector antenna. The AWG is controlled with a laptop through a MATLAB graphical user interface (GUI), and is capable of loading MATLAB data files containing arbitrary waveforms. On the receive side, the RAIL system consists of an AirMax 2G-16-90 sector antenna mounted on a linear motor driven rail that is controlled with a laptop through a MATLAB



(a) Measured defocused BPA image.



(b) Autofocused BPA image.

Fig. 7. Measured and autofocused BPA images for the experimental setup in Fig. 6.

GUI. The receive antenna is fed into a Tektronix TDS6124 digital storage oscilloscope, which records and stores data for post-processing.

Figure 7a depicts the measured defocused SAR image, where the defocus is primarily a result of unknown, random timing delays in the software triggers of the system which contribute to receiver position uncertainty during the collection. From Fig. 7a a near-field response is evidenced by the curvature of the energy in the image along the cross-range dimension. Hence, the PGA algorithm is not well-suited for this collection geometry as the far-field approximation does not hold (while the GPGA algorithm does not suffer from this same issue). Figure 7b depicts the autofocused SAR image after the third iteration of the GPGA algorithm, as well as the scatterers selected during the third iteration of the algorithm. The autofocused image more clearly displays the four plate reflectivities that are not discernible in the measured defocused image.

C. Gotcha SAR Data

In this section, the public release data set, referred to as “Gotcha Volumetric SAR Data Set, Version 1.0,” is used to produce example autofocus results using the GPGA algorithm [60]. The Gotcha data set consists of X-band SAR data with a bandwidth of 640 MHz [60]. For results shown herein, an aperture consisting of 469 slow-time pulses of data, ranging from 0° to 4° azimuth, is used.

1) *Using the PFA:* For applying the GPGA algorithm with the PFA, the Gotcha SAR data is first resampled from a polar

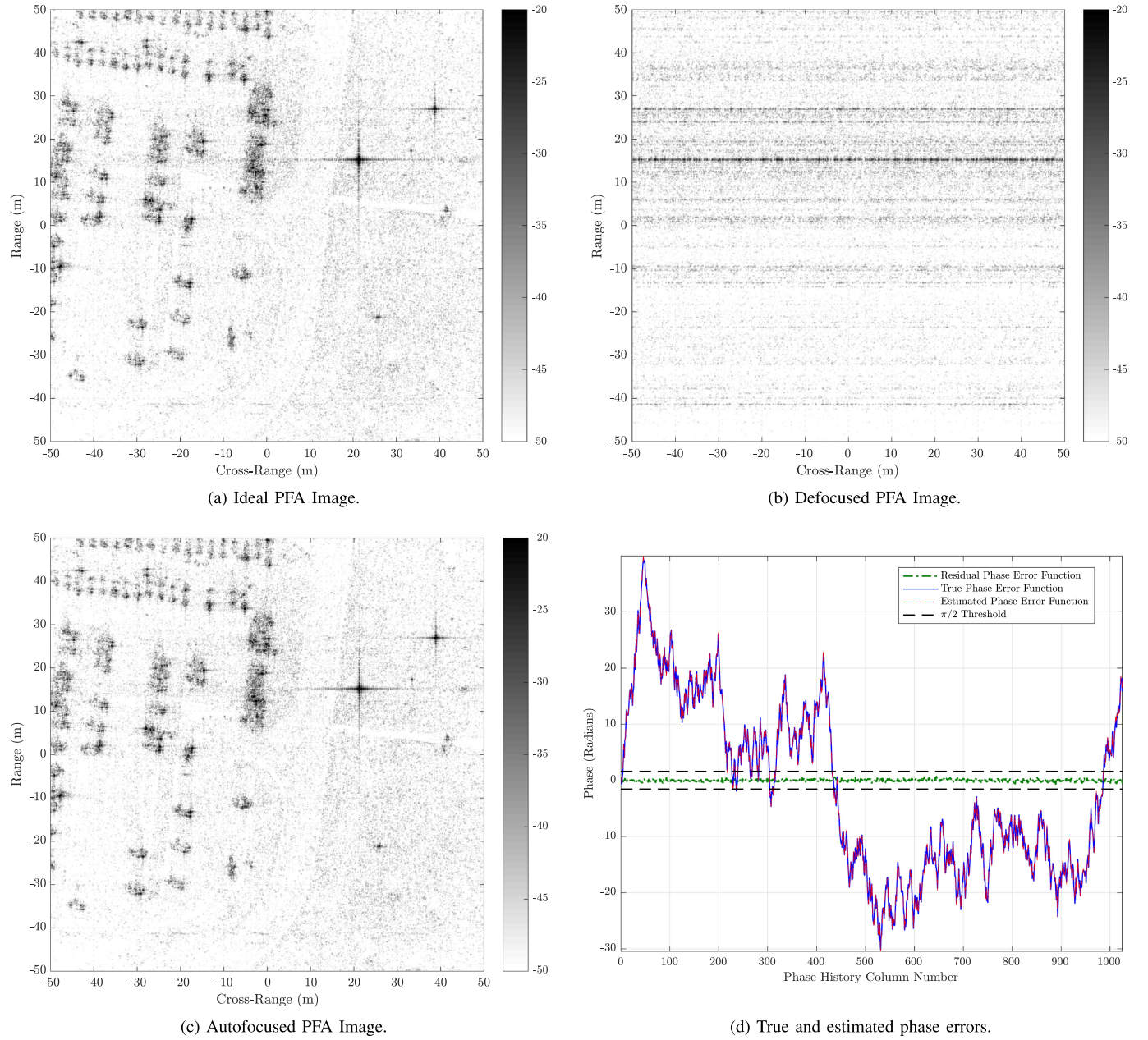


Fig. 8. GPGA algorithm autofocus results using the PFA for SAR image formation.

to a uniformly sampled Cartesian grid, resulting in 1024 slow-time pulses of resampled data. Then, synthetic phase errors are applied to the resampled data. The synthetic phase errors consist of white phase errors for each slow-time pulse chosen from a uniform distribution over the interval $[-\pi, \pi)$.

Figures 8a and 8b depict the ideal and defocused PFA SAR images. Figure 8c shows the autofocused PFA SAR image after applying the GPGA algorithm for three iterations. Figure 8d shows the true, estimated, and residual phase errors. For the shown estimated phase errors in Fig. 8d, quantities which the autofocus solution is invariant to are removed. These invariant quantities include: 1) constant phase offset, 2) 2π jumps in phase, and 3) linear phase ramp (which results in image translation) [65]. As evidenced by Figs. 8c and 8d, the GPGA algorithm

using the PFA is able to restore the defocused image to closely resemble the ideal image.

To provide additional insight into how the performance of the GPGA algorithm may change based on its configuration, the MSE of the argument of phase error estimates produced from the defocused SAR image in Fig. 8b is evaluated versus different configuration parameters. In particular, Table I is a tabulation of the MSE of the argument of phase error estimates and number of selected scatterers versus the applied relative threshold for scatterer selection and number of GPGA iterations run. As illustrated with the results in Table I, the MSE of the argument of phase error estimates is generally expected to decrease with the number of moderate or constant SINR scatterers selected and number of GPGA iterations run.

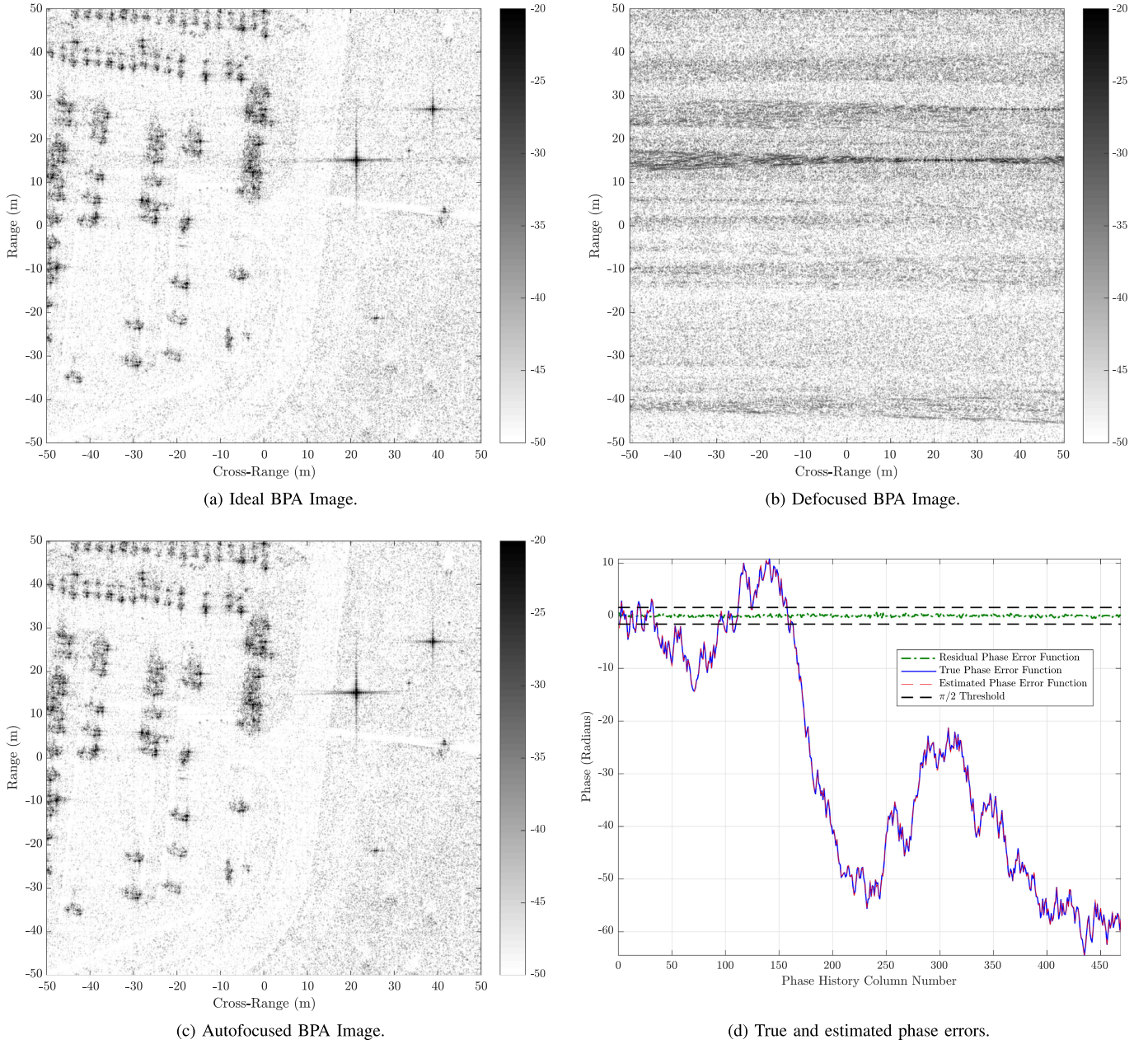


Fig. 9. GPGA algorithm autofocus results using the BPA for SAR image formation.

2) *Using the BPA*: For applying the GPGA algorithm with the BPA, synthetic phase errors are applied directly to the 469 pulses of data (i.e., there is no polar-to-Cartesian resampling of the data). Again, the synthetic phase errors consist of white phase errors for each slow-time pulse chosen from a uniform distribution over the interval $[-\pi, \pi)$.

Figures 9a and 9b depict the ideal and defocused BPA SAR images. Figure 9c shows the autofocused BPA SAR image after applying the GPGA algorithm for three iterations. Figure 9d shows the true, estimated, and residual phase errors. The true phase errors depicted in Fig. 9d differ from those shown in Fig. 8d as the phase errors for the BPA result are applied directly to the 469 pulses of data (versus being applied to the polar-to-Cartesian resampled data as in

Section VI-C1). Hence, the defocused image in Fig. 9b also differs from Fig. 8b due to the difference in the applied synthetic phase errors. For the estimated phase errors shown in Fig. 9d, quantities which the autofocus solution are invariant to are removed. These invariant quantities include: 1) constant phase offset and 2) 2π jumps in phase [65]. As evidenced by Figs. 9c and 9d, the GPGA algorithm using the BPA is able to restore the defocused image to closely resemble the ideal image.

Again, to provide additional insight into how the performance of the GPGA algorithm may change based on its configuration, the MSE of the argument of phase error estimates produced from the defocused SAR image in Fig. 9b is evaluated versus different configuration parameters. In particular, Table II is a

TABLE I
MSE OF THE ARGUMENT OF PHASE ERROR ESTIMATES IN RADIANS² AND
NUMBER OF SELECTED SCATTERERS VERSUS THE RELATIVE THRESHOLD FOR
SCATTERER SELECTION AND THE NUMBER OF GPGA ITERATIONS RUN

		Iteration			
		Iter. #1	Iter. #2	Iter. #3	Iter. #4
Rel. Threshold	15 dB	3.311 30	0.115 30	0.045 7	0.045 7
	10 dB	3.311 30	2.798 10	0.045 3	0.045 3
	5 dB	3.311 30	3.248 3	0.097 1	0.087 1

TABLE II
MSE OF THE ARGUMENT OF PHASE ERROR ESTIMATES IN RADIANS² AND
NUMBER OF SELECTED SCATTERERS VERSUS THE RELATIVE THRESHOLD FOR
SCATTERER SELECTION AND THE NUMBER OF GPGA ITERATIONS RUN

		Iteration			
		Iter. #1	Iter. #2	Iter. #3	Iter. #4
Rel. Threshold	15 dB	0.069 30	0.031 30	0.031 30	0.031 30
	10 dB	0.069 30	0.029 19	0.029 19	0.029 18
	5 dB	0.069 30	0.030 6	0.030 6	0.030 6

tabulation of the MSE of the argument of phase error estimates and number of selected scatterers versus the applied relative threshold for scatterer selection and number of GPGA iterations run. As illustrated with the results in Table II, the MSE of the argument of phase error estimates is generally expected to decrease with the number of moderate or constant SINR scatterers selected and number of GPGA iterations run.

VII. CONCLUSION

In this work, a GPGA algorithm was developed, which is applicable with both the PFA and BPA. The GPGA algorithm was shown to preserve the four crucial signal processing steps comprising the PGA algorithm, while alleviating the constraint of using a single scatterer per range cut for phase error estimation that exists with the PGA algorithm. Moreover, it was shown that the GPGA algorithm, whether using the PFA or BPA, provides an approximate MMLE of phase errors having marginalized over unknown complex-valued reflectivities of selected scatterers. Simulation and experimental results produced by applying the GPGA algorithm with the PFA and BPA were presented.

APPENDIX

A. Marginal Likelihood Derivation

Assume N complex-valued observations, $\mathbf{y} \in \mathbb{C}^N$, are made of an affine data model

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{b}, \quad (40)$$

where $\mathbf{A} \in \mathbb{C}^{N \times N}$, $\mathbf{x}, \mathbf{b} \in \mathbb{C}^N$, and each of the parameters is assumed to be independent with respect to one another. Moreover, assume $\mathbf{A}$ is arbitrarily distributed and the remaining parameters in the data model are distributed as

$$\mathbf{x} \sim \mathcal{CN}(\mathbf{0}, \mathbf{C}_x(\eta)), \quad (41)$$

$$\mathbf{b} \sim \mathcal{CN}(\boldsymbol{\mu}_b(\eta), \mathbf{C}_b(\eta)), \quad (42)$$

where $\boldsymbol{\mu}_b : \mathbb{C}^P \rightarrow \mathbb{C}^N$ returns the mean vector for $\mathbf{b}$, $\mathbf{C}_x, \mathbf{C}_b : \mathbb{C}^P \rightarrow \mathbb{C}^{N \times N}$ return the covariance matrix for $\mathbf{x}$ and $\mathbf{b}$, respectively, and $\eta \in \mathbb{C}^P$ is a vector containing all hyperparameters (i.e., distribution parameters). Given the descriptions above, the data observations, given all parameters except $\mathbf{b}$, are distributed as

$$\mathbf{y} | \mathbf{A}, \mathbf{x}, \eta \sim \mathcal{CN}(\boldsymbol{\mu}_y(\mathbf{A}\mathbf{x}, \eta), \mathbf{C}_b(\eta)), \quad (43)$$

(1)

where $\boldsymbol{\mu}_y : \mathbb{C}^N \times \mathbb{C}^P \rightarrow \mathbb{C}^N$, $\boldsymbol{\mu}_y(\mathbf{A}\mathbf{x}, \eta) = \mathbf{A}\mathbf{x} + \boldsymbol{\mu}_b(\eta)$. In the proceeding derivation it is shown that the marginal likelihood of $\mathbf{y}$, marginalizing over $\mathbf{x}$, is distributed as

$$\mathbf{y} | \mathbf{A}, \eta \sim \mathcal{CN}(\boldsymbol{\mu}_b(\eta), \mathbf{A}\mathbf{C}_x(\eta)\mathbf{A}^H + \mathbf{C}_b(\eta)), \quad (44)$$

assuming both $\mathbf{C}_x(\eta)$ and $\mathbf{C}_b(\eta)$ are full rank (hence, invertible).

The marginal likelihood, marginalizing over $\mathbf{x}$, is

$$\begin{aligned} \Pr(\mathbf{y} | \mathbf{A}, \eta) &= \int \Pr(\mathbf{y} | \mathbf{A}, \mathbf{x}, \eta) \Pr(\mathbf{x} | \mathbf{A}, \eta) d\mathbf{x} \\ &= \int \Pr(\mathbf{y} | \mathbf{A}, \mathbf{x}, \eta) \Pr(\mathbf{x}) d\mathbf{x}, \end{aligned} \quad (45)$$

recalling each of the parameters are assumed to be independent with respect to one another. Using Eqs. (41) and (43), and letting $\mathbf{z} : \mathbb{C}^N \times \mathbb{C}^P \rightarrow \mathbb{C}^N$, $\mathbf{z}(\mathbf{y}, \eta) = \mathbf{y} - \boldsymbol{\mu}_b(\eta)$, the marginal likelihood is (omitting arguments to functions to simplify notation)

$$\begin{aligned} \Pr(\mathbf{y} | \mathbf{A}, \eta) &= \int \frac{1}{\pi^N |\mathbf{C}_b|} e^{-(\mathbf{z} - \mathbf{A}\mathbf{x})^H \mathbf{C}_b^{-1} (\mathbf{z} - \mathbf{A}\mathbf{x})} \\ &\quad \frac{1}{\pi^N |\mathbf{C}_x|} e^{-\mathbf{x}^H \mathbf{C}_x^{-1} \mathbf{x}} d\mathbf{x} \\ &= \frac{e^{-\mathbf{z}^H \mathbf{C}_b^{-1} \mathbf{z}}}{\pi^N |\mathbf{C}_b| |\mathbf{C}_x|} \int \frac{1}{\pi^N} e^{-\mathbf{x}^H (\mathbf{C}_x^{-1} + \mathbf{A}^H \mathbf{C}_b^{-1} \mathbf{A}) \mathbf{x}} \\ &\quad e^{2\text{Re}\{\mathbf{z}^H \mathbf{C}_b^{-1} \mathbf{A}\mathbf{x}\}} d\mathbf{x}. \end{aligned} \quad (46)$$

Letting $\mathbf{C}_1^{-1}(\eta) = \mathbf{C}_x^{-1}(\eta) + \mathbf{A}^H \mathbf{C}_b^{-1}(\eta) \mathbf{A}$ and $\tilde{\mathbf{y}}(\eta) = \mathbf{A}^H \mathbf{C}_b^{-1}(\eta) \mathbf{z}$, the result in Eq. (46) becomes

$$\begin{aligned} \Pr(\mathbf{y} | \mathbf{A}, \eta) &= \frac{|\mathbf{C}_1|}{\pi^N |\mathbf{C}_b| |\mathbf{C}_x|} e^{-\mathbf{z}^H \mathbf{C}_b^{-1} \mathbf{z}} e^{\tilde{\mathbf{y}}^H \mathbf{C}_1 \tilde{\mathbf{y}}} \\ &\quad \int \frac{1}{\pi^N |\mathbf{C}_1|} e^{-(\mathbf{x} - \mathbf{C}_1 \tilde{\mathbf{y}})^H \mathbf{C}_1^{-1} (\mathbf{x} - \mathbf{C}_1 \tilde{\mathbf{y}})} d\mathbf{x} \\ &= \frac{|\mathbf{C}_1|}{\pi^N |\mathbf{C}_b| |\mathbf{C}_x|} e^{-\mathbf{z}^H \mathbf{C}_b^{-1} \mathbf{z}} e^{\tilde{\mathbf{y}}^H \mathbf{C}_1 \tilde{\mathbf{y}}}. \end{aligned} \quad (47)$$

Substituting back in for $\tilde{\mathbf{y}}$ and letting $\mathbf{C}_2^{-1}(\boldsymbol{\eta}) = \mathbf{C}_b^{-1}(\boldsymbol{\eta}) - \mathbf{C}_b^{-1}(\boldsymbol{\eta}) \mathbf{A} \mathbf{C}_1(\boldsymbol{\eta}) \mathbf{A}^H \mathbf{C}_b^{-H}(\boldsymbol{\eta})$ produces

$$\begin{aligned} \Pr(\mathbf{y} \mid \mathbf{A}, \boldsymbol{\eta}) &= \frac{|\mathbf{C}_1|}{\pi^N |\mathbf{C}_b| |\mathbf{C}_x|} e^{-\mathbf{z}^H \mathbf{C}_b^{-1} \mathbf{z}} e^{\mathbf{z}^H \mathbf{C}_b^{-1} \mathbf{A} \mathbf{C}_1 \mathbf{A}^H \mathbf{C}_b^{-H} \mathbf{z}} \\ &= \frac{1}{\pi^N |\mathbf{C}_b| |\mathbf{C}_x| |\mathbf{C}_1^{-1}|} e^{-\mathbf{z}^H \mathbf{C}_2^{-1} \mathbf{z}}. \end{aligned} \quad (48)$$

Using the Woodbury matrix identity it is seen that

$$\mathbf{C}_2^{-1}(\boldsymbol{\eta}) = (\mathbf{A} \mathbf{C}_x(\boldsymbol{\eta}) \mathbf{A}^H + \mathbf{C}_b(\boldsymbol{\eta}))^{-1}, \quad (49)$$

implying $\mathbf{C}_2(\boldsymbol{\eta}) = \mathbf{A} \mathbf{C}_x(\boldsymbol{\eta}) \mathbf{A}^H + \mathbf{C}_b(\boldsymbol{\eta})$. Using the matrix determinant lemma it is seen that

$$|\mathbf{C}_2(\boldsymbol{\eta})| = |\mathbf{C}_1^{-1}(\boldsymbol{\eta})| |\mathbf{C}_x(\boldsymbol{\eta})| |\mathbf{C}_b(\boldsymbol{\eta})|. \quad (50)$$

Using Eqs. (49) and (50), and substituting in the definition for $\mathbf{z}$ in Eq. (48) produces

$$\begin{aligned} \Pr(\mathbf{y} \mid \mathbf{A}, \boldsymbol{\eta}) &= \frac{1}{\pi^N |\mathbf{C}_2|} e^{-(\mathbf{y} - \boldsymbol{\mu}_b)^H \mathbf{C}_2^{-1} (\mathbf{y} - \boldsymbol{\mu}_b)} \\ &= \mathcal{CN}(\boldsymbol{\mu}_b, \mathbf{A} \mathbf{C}_x \mathbf{A}^H + \mathbf{C}_b), \end{aligned} \quad (51)$$

which is equivalent to the marginal likelihood in Eq. (44).

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REFERENCES

- [1] W. G. Carrara, R. S. Goodman, and R. M. Majewski, *Spotlight Synthetic Aperture Radar Signal Processing Algorithms*. Boston, MA, USA: Artech House, 1995.
- [2] C. V. Jakowatz Jr., D. E. Wahl, P. H. Eichel, D. C. Ghiglia, and P. A. Thompson, *Spotlight-Mode Synthetic Aperture Radar: A Signal Processing Approach*. Boston, MA, USA: Kluwer, 1996.
- [3] R. M. Naething and R. D. West, "Analysis of SAR autofocus performance," *Proc. SPIE*, vol. 8714, 2013, Art. no. 871414.
- [4] R. G. Paxman and J. C. Marron, "Aberration correction of speckled imagery with an image-sharpness criterion," *Proc. SPIE*, vol. 976, pp. 37–47, 1988.
- [5] P. H. Eichel, D. C. Ghiglia, and C. V. Jakowatz Jr., "Speckle processing method for synthetic-aperture-radar phase correction," *Opt. Lett.*, vol. 14, no. 1, pp. 1–3, 1989.
- [6] P. H. Eichel and C. V. Jakowatz Jr., "Phase-gradient algorithm as an optimal estimator of the phase derivative," *Opt. Lett.*, vol. 14, no. 20, pp. 1101–1103, 1989.
- [7] J. C. Curlander and R. N. McDonough, *Synthetic Aperture Radar Systems and Signal Processing*. New York, NY, USA: Wiley, 1991.
- [8] C. V. Jakowatz Jr. and D. E. Wahl, "Eigenvector method for maximum-likelihood estimation of phase errors in synthetic-aperture-radar imagery," *J. Opt. Soc. Amer.*, vol. 10, no. 12, pp. 2539–2546, 1993.
- [9] D. E. Wahl, P. H. Eichel, D. C. Ghiglia, and C. V. Jakowatz Jr., "Phase gradient autofocus—a robust tool for high resolution SAR phase correction," *IEEE Trans. Aerosp. Electron. Syst.*, vol. AES-30, no. 3, pp. 827–835, Jul. 1994.
- [10] D. E. Wahl, C. V. Jakowatz Jr., P. A. Thompson, and D. C. Ghiglia, "New approach to strip-map SAR autofocus," in *Proc. IEEE 6th Digit. Signal Process. Workshop*, 1994, pp. 53–56.
- [11] J. S. Bates, "The phase gradient autofocus algorithm with range dependent stripmap SAR," Ph.D. dissertation, Dept. Elect. Comput. Eng., Brigham Young Univ., Provo, UT, USA, 1998.
- [12] H. L. Chan and T. S. Yeo, "Noniterative quality phase-gradient autofocus (QPGA) algorithm for spotlight SAR imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 36, no. 5, pp. 1531–1539, Sep. 1998.
- [13] D. G. Thompson, J. S. Bates, D. V. Arnold, D. G. Long, and A. Robertson, "Range dependent phase gradient autofocus," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, 1998, vol. 5, pp. 2634–2636.
- [14] J. R. Fienup, "Synthetic-aperture radar autofocus by maximizing sharpness," *Opt. Lett.*, vol. 25, no. 4, pp. 221–223, 2000.
- [15] J. R. Fienup and J. J. Miller, "Generalized image sharpness metrics for correcting phase errors," in *Proc. J. Signal Recovery Syn.*, 2001, Art. no. STuD1. [Online]. Available: <http://www.osapublishing.org/abstract.cfm?URI=SRS-2001-STuD1>
- [16] H. J. Callow, M. P. Hayes, and P. T. Gough, "Stripmap phase gradient autofocus," in *Proc. Oceans Celebrating Past ... Teaming Toward Future*, 2003, vol. 5, pp. 2414–2421.
- [17] J. R. Fienup and J. J. Miller, "Aberration correction by maximizing generalized sharpness metrics," *J. Opt. Soc. Amer.*, vol. 20, no. 4, pp. 609–620, 2003.
- [18] J. Kolman, "PACE: An autofocus algorithm for SAR," in *Proc. IEEE Int. Radar Conf.*, 2005, pp. 310–314.
- [19] R. L. Morrison and M. N. Do, "A multichannel approach to metric-based SAR autofocus," in *Proc. IEEE Int. Conf. Image Process.*, 2005, vol. 2, pp. 1067–1070.
- [20] T. J. Kragh, "Monotonic iterative algorithm for minimum-entropy autofocus," in *Proc. Adapt. Sensor Array Process. Workshop*, 2006, pp. 645–648.
- [21] R. L. Morrison and M. N. Do, "Multichannel autofocus algorithm for synthetic aperture radar," in *Proc. IEEE Int. Conf. Image Process.*, 2006, pp. 2341–2344.
- [22] B. D. Rigling and R. L. Moses, "Motion measurement errors and autofocus in bistatic SAR," *IEEE Trans. Image Process.*, vol. 15, no. 4, pp. 1008–1016, Apr. 2006.
- [23] W. L. Van Rossum, M. P. G. Otten, and R. J. P. Van Bree, "Extended PGA for range migration algorithms," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 42, no. 2, pp. 478–488, Apr. 2006.
- [24] M. Martorella, M. Preiss, B. Haywood, and B. Bates, "Polarimetric phase gradient autofocus," in *Proc. Int. Geosci. Remote Sens. Symp.*, 2007, pp. 133–136.
- [25] R. L. Morrison, M. N. Do, and D. C. Munson, "SAR image autofocus by sharpness optimization: A theoretical study," *IEEE Trans. Image Process.*, vol. 16, no. 9, pp. 2309–2321, Sep. 2007.
- [26] T. J. Schulz, "Optimal sharpness function for SAR autofocus," *IEEE Signal Process. Lett.*, vol. 14, no. 1, pp. 27–30, Jan. 2007.
- [27] H. J. Cho and D. C. Munson, "Overcoming polar-format issues in multichannel SAR autofocus," in *Proc. Asilomar Conf. Signals, Syst. Comput.*, 2008, pp. 523–527.
- [28] C. V. Jakowatz Jr., D. E. Wahl, and D. A. Yocky, "Beamforming as a foundation for spotlight-mode SAR image formation by backprojection," *Proc. SPIE*, vol. 6970, 2008, Art. no. 697015.
- [29] K.-H. Liu and D. C. Munson, "Fourier-domain multichannel autofocus for synthetic aperture radar," in *Proc. Asilomar Conf. Signals, Syst. Comput.*, 2008, pp. 848–852.
- [30] K.-H. Liu and D. C. Munson Jr., "Autofocus in multistatic passive SAR imaging," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process.*, 2008, pp. 1277–1280.
- [31] C. V. Jakowatz Jr. and D. E. Wahl, "Considerations for autofocus of spotlight-mode SAR imagery created using a beamforming algorithm," *Proc. SPIE*, vol. 7337, 2009, Art. no. 73379.
- [32] R. L. Morrison, M. N. Do, and D. C. Munson, "MCA: A multichannel approach to SAR autofocus," *IEEE Trans. Image Process.*, vol. 18, no. 4, pp. 840–853, Apr. 2009.
- [33] K.-H. Liu, A. Wiesel, and D. C. Munson, "Synthetic aperture radar autofocus via semidefinite relaxation," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process.*, 2010, pp. 1342–1345.
- [34] H. J. Cho and D. C. Munson Jr., "CBP-based multichannel autofocus for near-field SAR imaging," *Proc. SPIE*, vol. 8051, 2011, Art. no. 80519.
- [35] K.-H. Liu and D. C. Munson, "Fourier-domain multichannel autofocus for synthetic aperture radar," *IEEE Trans. Image Process.*, vol. 20, no. 12, pp. 3544–3552, Dec. 2011.
- [36] K.-H. Liu, A. Wiesel, and D. C. Munson, "Maximum likelihood SAR autofocus with low-return region," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process.*, 2011, pp. 1377–1380.
- [37] J. N. Ash, "An autofocus method for backprojection imagery in synthetic aperture radar," *IEEE Geosci. Remote Sens. Lett.*, vol. 9, no. 1, pp. 104–108, Jan. 2012.

- [38] K.-H. Liu, A. Wiesel, and D. C. Munson, "Synthetic aperture radar autofocus based on a bilinear model," *IEEE Trans. Image Process.*, vol. 21, no. 5, pp. 2735–2746, May 2012.
- [39] R. D. West, J. Gunther, and T. Moon, "Convex optimization of subspace fitting autofocus for stripmap SAR images," in *Proc. IEEE Radar Conf.*, 2012, pp. 731–734.
- [40] S. Zhou, M. Xing, X. G. Xia, Y. Li, L. Zhang, and Z. Bao, "An azimuth-dependent phase gradient autofocus (APGA) algorithm for airborne/stationary BiSAR imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 6, pp. 1290–1294, Nov. 2013.
- [41] M. I. Duersch, "Backprojection for synthetic aperture radar," Ph.D. dissertation, Dept. Elect. Comput. Eng., Brigham Young Univ., Provo, UT, USA, 2013.
- [42] K.-H. Liu, A. Wiesel, and D. C. Munson, "Synthetic aperture radar autofocus via semidefinite relaxation," *IEEE Trans. Image Process.*, vol. 22, no. 6, pp. 2317–2326, Jun. 2013.
- [43] K. Hu, X. Zhang, S. He, H. Zhao, and J. Shi, "A less-memory and high-efficiency autofocus back projection algorithm for SAR imaging," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 4, pp. 890–894, Apr. 2015.
- [44] S. Wei, X. Zhang, K. Hu, and W. Wu, "LASAR autofocus imaging using maximum sharpness back projection via semidefinite programming," in *Proc. IEEE Radar Conf.*, 2015, pp. 1311–1315.
- [45] S. Chen, F. Lu, J. Wang, and M. Liu, "An improved phase gradient autofocus method for one-stationary bistatic SAR," in *Proc. IEEE Int. Conf. Signal Process., Commun., Comput. Conf.*, 2016, pp. 1–5.
- [46] H. J. Cho, "Autofocus and back-projection in synthetic aperture radar imaging," Ph.D. dissertation, Dept., Elect., Eng., Univ. Michigan, Ann Arbor, MI, USA, 2016.
- [47] A. Evers, N. Zimmerman, and J. A. Jackson, "Semidefinite relaxation autofocus for bistatic backprojection SAR," in *Proc. IEEE Radar Conf.*, 2017, pp. 1332–1337.
- [48] Y. Li *et al.*, "An autofocus method based on maximum image sharpness for fast factorized back-projection," in *Proc. IEEE Radar Conf.*, 2017, pp. 1201–1204.
- [49] X. Wang, Q. Wu, and J. Yang, "Extended PGA processing of high resolution airborne SAR imagery reconstructed via backprojection algorithm," in *Proc. IEEE Int. Radar Conf.*, 2016, pp. 1–3.
- [50] S. Wei, L. Zhou, X. Zhang, and J. Shi, "Fast back-projection autofocus for linear array SAR 3-D imaging via maximum sharpness," in *Proc. IEEE Radar Conf.*, 2018, pp. 525–530.
- [51] A. Evers and J. A. Jackson, "SAR scene size limits for spatially invariant defocus with near- and far-field geometries," *IEEE Trans. Comput. Imag.*, vol. 4, no. 4, pp. 640–647, Dec. 2018.
- [52] R. Zadeh, "Lecture notes on singular value decomposition," 2017. [Online]. Available: <https://stanford.edu/rezab/classes/cme323/S17/notes/lecture17/cme323 lec17.pdf>
- [53] M. Weiss, "Analysis of Some modified cell-averaging CFAR processors in multiple-target situations," *IEEE Trans. Aerosp. Electron. Syst.*, vol. AES-18, no. 1, pp. 102–114, Jan. 1982.
- [54] H. Rohling, "Radar CFAR thresholding in clutter and multiple target situations," *IEEE Trans. Aerosp. Electron. Syst.*, vol. AES-19, no. 4, pp. 608–621, Jul. 1983.
- [55] P. P. Gandhi and S. A. Kassam, "Analysis of CFAR processors in non-homogeneous background," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 24, no. 4, pp. 427–445, Jul. 1988.
- [56] N. Levanon, "Detection loss due to interfering targets in ordered statistics CFAR," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 24, no. 6, pp. 678–681, Nov. 1988.
- [57] Z.-Q. Luo, W.-K. Ma, A. M.-C. So, Y. Ye, and S. Zhang, "Semidefinite relaxation of quadratic optimization problems," *IEEE Signal Process. Mag.*, vol. 27, no. 3, pp. 20–34, May 2010.
- [58] J. B. Keller, "Geometrical theory of diffraction," *J. Opt. Soc. Amer.*, vol. 52, no. 2, pp. 116–130, 1962.
- [59] L. C. Potter and R. L. Moses, "Attributed scattering centers for SAR ATR," *IEEE Trans. Image Process.*, vol. 6, no. 1, pp. 79–91, Jan. 1997.
- [60] C. H. Casteel Jr., L. A. Gorham, M. J. Minardi, S. M. Scarborough, K. D. Naidu, and U. K. Majumder, "A challenge problem for 2D/3D imaging of targets from a volumetric data set in an urban environment," *Proc. SPIE*, vol. 6568, 2007, Art. no. 65687.
- [61] J. R. Gutierrez, "Passive synthetic aperture radar imaging using commercial OFDM communication networks," Ph.D. dissertation, Dept. Air Force Air Univ., Air Force Inst. Technol., Wright-Patterson Air Force Base, OH, USA, 2012.
- [62] J. R. Gutierrez and J. A. Jackson, "Collecting and processing WiMAX ground returns for SAR imaging," in *Proc. IEEE Radar Conf.*, 2013, pp. 1–6.
- [63] J. R. Gutierrez, J. A. Jackson, and M. A. Temple, "Receive signal processing for OFDM-based radar imaging," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process.*, 2013, pp. 2775–2779.
- [64] A. Evers and J. A. Jackson, "Experimental passive SAR imaging exploiting LTE, DVB, and DAB signals," in *Proc. IEEE Radar Conf.*, 2014, pp. 680–685.
- [65] J. R. Fienup, "Invariant error metrics for image reconstruction," *Appl. Opt.*, vol. 36, no. 32, pp. 8352–8357, 1997.



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